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name: <unnamed>
log: E:\OneDrive - UNBC\AP_UNBC\ECON712_Fall_2024\Problem Set 2\STATA_Solutions\Titanic_log.smcl
log type: smcl
opened on: 9 Dec 2024, 16:19:36

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1 .
2 . *****
3 . *****
4 . ***** Question II, part a
5 . *****
6 . *****
7 . use "titanic",clear
   (Population at Risk and Death Rates for an Unusual Episode)

8 . gen female=(sex==0)

9 . label variable female "Female"

10 . gen male=(sex==1)

11 . label variable male "Male"

12 . gen      s=1 if (female==1 & age==1)
   (1,776 missing values generated)

13 . replace s=2 if (female==1 & age==0)
   (45 real changes made)

14 . replace s=3 if (female==0 & age==1)
   (1,667 real changes made)

15 . replace s=4 if (female==0 & age==0)
   (64 real changes made)

16 . gen      d=1 if class==1
   (1,876 missing values generated)

17 . replace d=0 if class!=1
   (1,876 real changes made)

18 . summarize survived if d==1

```

Variable	Obs	Mean	Std. dev.	Min	Max
survived	325	.6246154	.4849687	0	1

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19 . gen ey1=r(mean)
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20 . summarize survived if d==0
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Variable	Obs	Mean	Std. dev.	Min	Max
survived	1,876	.2707889	.4444858	0	1

21 . gen ey0=r(mean)

22 . gen sdo=ey1-ey0

23 . su sdo

Variable	Obs	Mean	Std. dev.	Min	Max
sdo	2,201	.3538265	0	.3538265	.3538265

24 .

25 . cap n drop ey1 ey0

26 . su survived if s==1 & d==1

Variable	Obs	Mean	Std. dev.	Min	Max
survived	144	.9722222	.1649091	0	1

27 . gen ey11=r(mean)

28 . label variable ey11 "Average survival for female adult in treatment"

29 . su survived if s==1 & d==0

Variable	Obs	Mean	Std. dev.	Min	Max
survived	281	.6263345	.4846395	0	1

30 . gen ey10=r(mean)

31 . label variable ey10 "Average survival for female adult in control"

32 . gen diff1=ey11-ey10

33 . label variable diff1 "Difference in survival for female adult"

34 . su survived if s==2 & d==1

Variable	Obs	Mean	Std. dev.	Min	Max
survived	1	1	.	1	1

35 . gen ey21=r(mean)

36 . su survived if s==2 & d==0

Variable	Obs	Mean	Std. dev.	Min	Max
survived	44	.6136364	.4925448	0	1

37 . gen ey20=r(mean)

38 . gen diff2=ey21-ey20

39 . su survived if s==3 & d==1

Variable	Obs	Mean	Std. dev.	Min	Max
survived	175	.3257143	.4699859	0	1

40 . gen ey31=r(mean)

41 . su survived if s==3 & d==0

Variable	Obs	Mean	Std. dev.	Min	Max
survived	1,492	.1883378	.3911128	0	1

42 . gen ey30=r(mean)

43 . gen diff3=ey31-ey30

44 . su survived if s==4 & d==1

Variable	Obs	Mean	Std. dev.	Min	Max
survived	5	1	0	1	1

45 . gen ey41=r(mean)

46 . su survived if s==4 & d==0

Variable	Obs	Mean	Std. dev.	Min	Max
survived	59	.4067797	.4954498	0	1

47 . gen ey40=r(mean)

48 . gen diff4=ey41-ey40

49 . count if s==1 & d==0
281

50 . count if s==2 & d==0
44

51 . count if s==3 & d==0
1,492

52 . count if s==4 & d==0
59

53 . count if d == 0
1,876

54 .

55 . gen wt1=425/2201

56 . gen wt2=45/2201

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57 . gen wt3=1667/2201
58 . gen wt4=64/2201
59 . gen wate=diff1*wt1 + diff2*wt2 + diff3*wt3 + diff4*wt4
60 . sum wate sdo

```

Variable	Obs	Mean	Std. dev.	Min	Max
wate	2,201	.1959842	0	.1959842	.1959842
sdo	2,201	.3538265	0	.3538265	.3538265

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61 .
62 . *****
63 . *****
64 . ***** Question II, part c
65 . *****
66 . *****
67 .
68 . reg survived d age female, robust

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Linear regression              Number of obs   =      2,201
                              F(3, 2197)       =      301.73
                              Prob > F         =      0.0000
                              R-squared        =      0.2388
                              Root MSE     =      .40836

```

survived	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
d	.2320018	.0258727	8.97	0.000	.1812644	.2827392
age	-.1355253	.0492844	-2.75	0.006	-.2321741	-.0388765
female	.4644836	.0241776	19.21	0.000	.4170703	.5118969
_cons	.3184058	.0488643	6.52	0.000	.2225807	.4142308

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69 .
70 .
71 . *****
72 . *****
73 . ***** Question II, part d
74 . *****
75 . *****
76 . gen first_female=d*female
77 .
78 . reg survived first_female d female age, robust

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```

Linear regression              Number of obs   =      2,201
                              F(4, 2196)       =      541.15
                              Prob > F         =      0.0000
                              R-squared        =      0.2449
                              Root MSE     =      .40683

```

survived	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
first_female	.2178787	.0477827	4.56	0.000	.1241747	.3115826
d	.1493489	.0364819	4.09	0.000	.0778062	.2208916
female	.4132482	.0292977	14.11	0.000	.3557941	.4707023
age	-.1512136	.0493999	-3.06	0.002	-.248089	-.0543383
_cons	.3421088	.0492363	6.95	0.000	.2455542	.4386634

79 .

80 .
end of do-file

81 . exit, clear